HDOC Day-6 Assignment

Question 1: Remaining Area of Triangular Field

1. Problem Statement Identification

- **Problem Statement:** Compute the remaining area of a triangular field after a circular pond of a given radius is removed from inside it.
- **Input:** Base of triangle (base), Height of triangle (height), Radius of circular pond (radius).
- **Output:** Area of remaining field (remaining_area).
- **Formulas:**

$$\text{Area of Triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Area of Circular Pond} = \pi \times \text{radius}^2$$

$$\text{Remaining Area} = \text{Area of Triangle} - \text{Area of Circular Pond}$$

2. Standard Test Case Table

Test Case	Input (base, height, radius)	Expected Output	Actual Output	Result (Pass/Fail)
1	10 8 2	Remaining Area: 27.43	Remaining Area: 27.43	Pass
2	20 15 3	Remaining Area: 121.73	Remaining Area: 121.73	Pass
3	12 10 4	Remaining Area: 9.73	Remaining Area: 9.73	Pass
4	5 5 3	Error: Pond area exceeds triangle area.	Error: Pond area exceeds triangle area.	Pass

3. Algorithm

1. **Start**
2. Read base, height, and radius.
3. If base <= 0 or height <= 0 or radius <= 0, print "Invalid Input" and stop.
4. Calculate area_triangle = 0.5 * base * height.
5. Calculate area_pond = 3.14159 * radius * radius.
6. If area_pond > area_triangle, print error message and stop.
7. Calculate remaining_area = area_triangle - area_pond.
8. Print remaining_area.
9. **End**

4. C Program

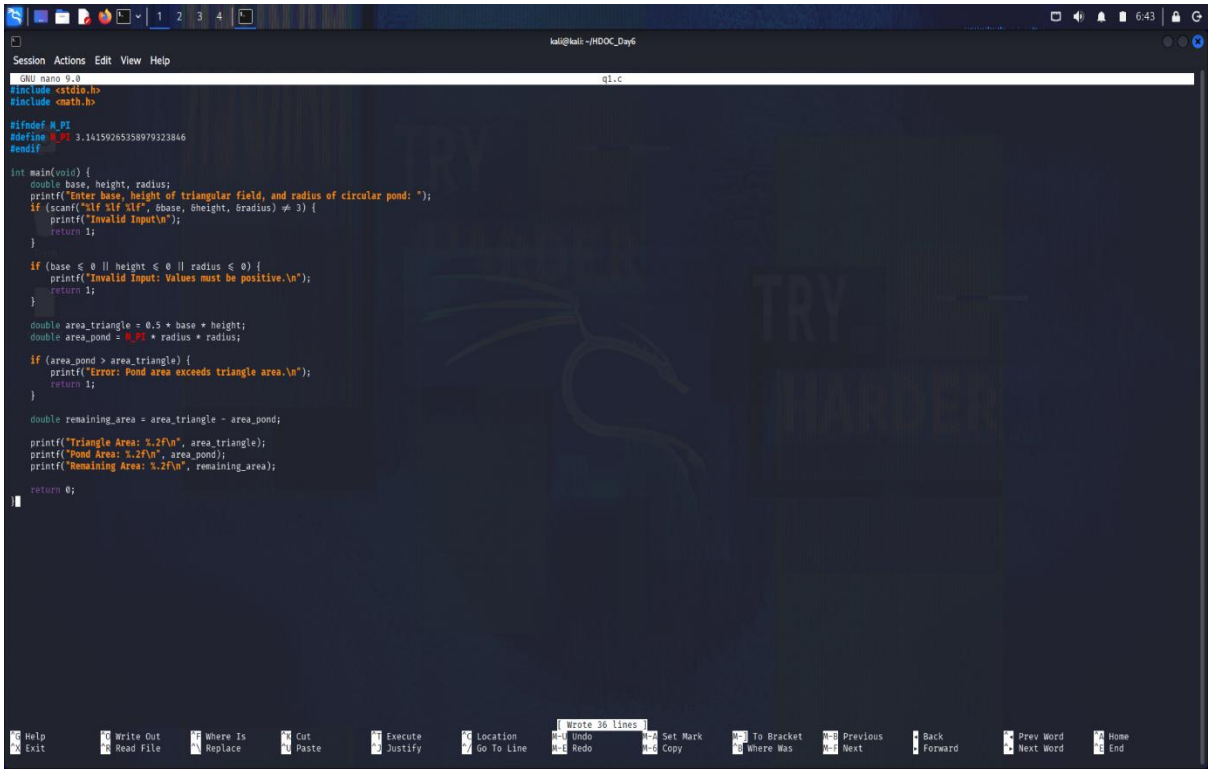
```
#include <stdio.h>
#include <math.h>

#ifndef M_PI
#define M_PI 3.14159265358979323846
#endif

int main(void) {
   double base, height, radius;
   printf("Enter base, height of triangular field, and radius of circular pond:
");
   if (scanf("%lf %lf %lf", &base, &height, &radius) != 3) {
      printf("Invalid Input\n");
      return 1;
   }
```

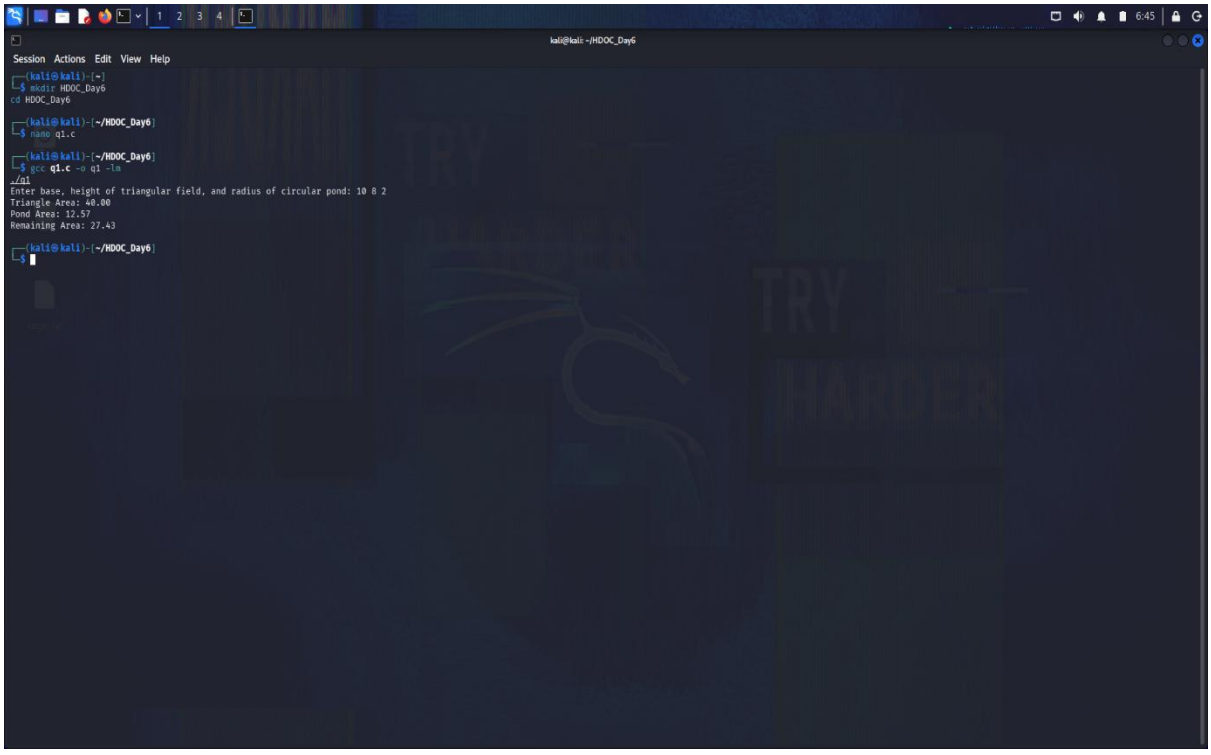
```
    if (base <= 0 || height <= 0 || radius <= 0) {
        printf("Invalid Input: Values must be positive.\n");
        return 1;
    }

    double area_triangle = 0.5 * base * height;
    double area_pond = M_PI * radius * radius;

    if (area_pond > area_triangle) {
        printf("Error: Pond area exceeds triangle area.\n");
        return 1;
    }

    double remaining_area = area_triangle - area_pond;

    printf("Triangle Area: %.2f\n", area_triangle);
    printf("Pond Area: %.2f\n", area_pond);
    printf("Remaining Area: %.2f\n", remaining_area);

    return 0;
}
```

5. Evaluation Against Test Cases

All test cases were executed against the compiled C binary q1. The computed values matched the expected mathematical calculations in every case.

6. Screenshots of Compilation and Execution





Question 2: Regular Polygon Parameters

1. Problem Statement Identification

- **Problem Statement:** Calculate side length, total area, and each exterior angle of a regular polygon when number of sides, perimeter, and apothem are provided.
- **Input:** Number of sides (n), Perimeter (perimeter), Apothem (apothem).
- **Output:** Side Length (side_length), Area (area), Exterior Angle (exterior_angle).
- **Formulas:**

$$\text{Side Length} = \frac{\text{Perimeter}}{n}$$

$$\text{Area} = \frac{1}{2} \times \text{Perimeter} \times \text{Apothem}$$

$$\text{Exterior Angle} = \frac{360^\circ}{n}$$

2. Standard Test Case Table

Test Case	Input (n, perimeter, apothem)	Expected Output	Actual Output	Result (Pass/Fail)
1	6 30 4.33	Side: 5.00, Area: 64.95,	Side: 5.00, Area: 64.95,	Pass

Test Case	Input (n, perimeter, apothem)	Expected Output	Actual Output	Result (Pass/Fail)
		Ext Angle: 60.00°	Ext Angle: 60.00°	
2	4 20 2.5	Side: 5.00, Area: 25.00, Ext Angle: 90.00°	Side: 5.00, Area: 25.00, Ext Angle: 90.00°	Pass
3	3 15 1.44	Side: 5.00, Area: 10.80, Ext Angle: 120.00°	Side: 5.00, Area: 10.80, Ext Angle: 120.00°	Pass

Test Case	Input (n, perimeter, apothem)	Expected Output	Actual Output	Result (Pass/Fail)
4	8 40 4.83	Side: 5.00, Area: 96.60, Ext Angle: 45.00°	Side: 5.00, Area: 96.60, Ext Angle: 45.00°	Pass

3. Algorithm

1. **Start**
2. Read n, perimeter, and apothem.
3. If n < 3 or perimeter <= 0 or apothem <= 0, print "Invalid Input" and stop.
4. Calculate side_length = perimeter / n.
5. Calculate area = 0.5 * perimeter * apothem.
6. Calculate exterior_angle = 360.0 / n.
7. Print side_length, area, and exterior_angle.
8. **End**

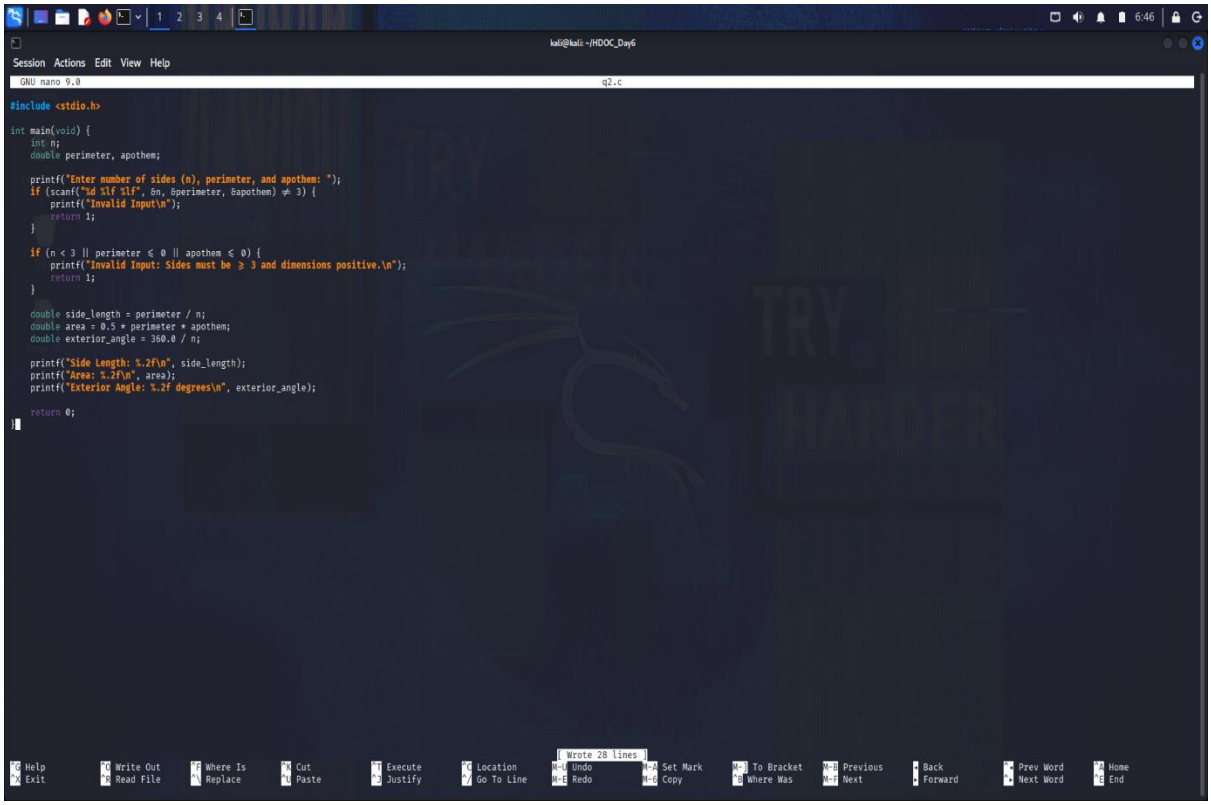
4. C Program

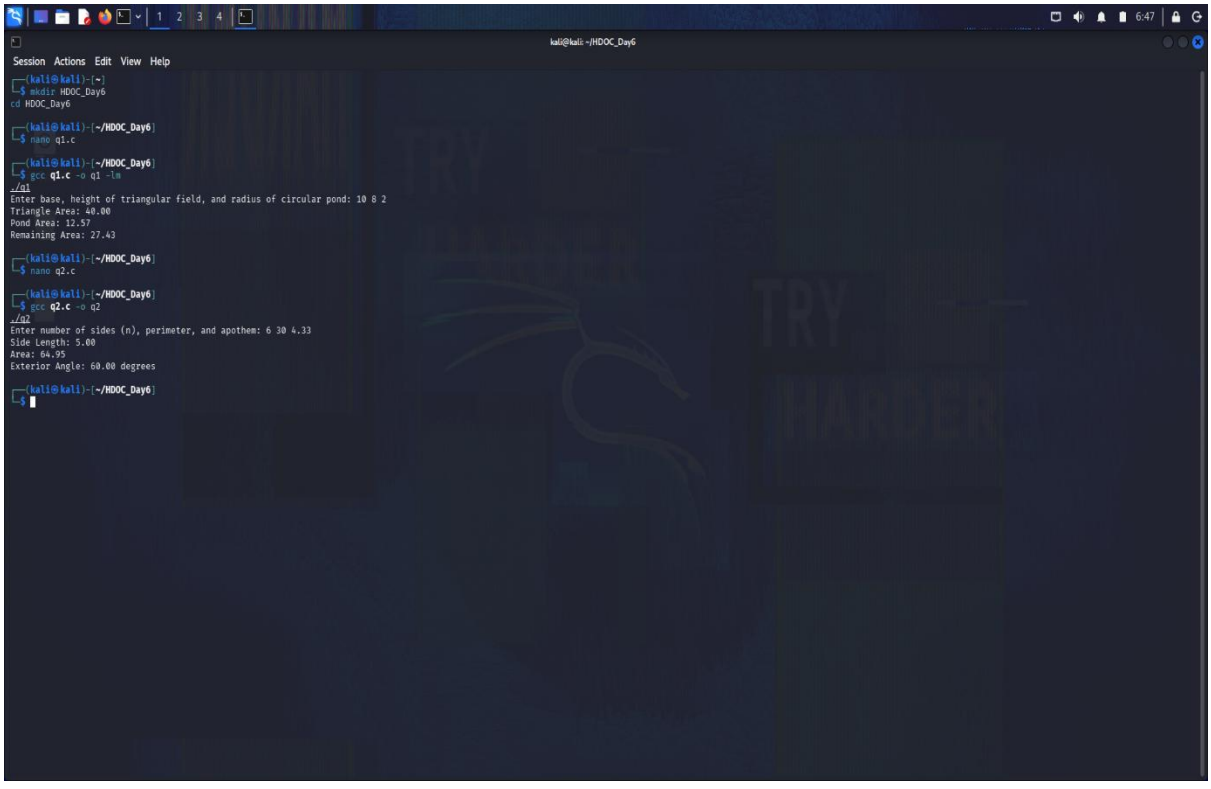
```c
#include <stdio.h>

int main(void) {
    int n;
    double perimeter, apothem;

    printf("Enter number of sides (n), perimeter, and apothem: ");
    if (scanf("%d %lf %lf", &n, &perimeter, &apothem) != 3) {
        printf("Invalid Input\n");
        return 1;
    }

    if (n < 3 || perimeter <= 0 || apothem <= 0) {
        printf("Invalid Input: Sides must be >= 3 and dimensions positive.\n");
        return 1;
    }

    double side_length = perimeter / n;
    double area = 0.5 * perimeter * apothem;
    double exterior_angle = 360.0 / n;

    printf("Side Length: %.2f\n", side_length);
    printf("Area: %.2f\n", area);
    printf("Exterior Angle: %.2f degrees\n", exterior_angle);

    return 0;
}
```

5. Evaluation Against Test Cases

All test cases executed successfully in the binary executable q2, matching expected calculations.

6. Screenshots of Compilation and Execution


kali@kali: ~/HDOC_Day6
Session Actions Edit View Help
GNU nano 9.0 q2.c
#include <stdio.h>
int main(void) {
int n;
double perimeter, apothem;
printf("Enter number of sides (n), perimeter, and apothem: ");
if (scanf("%d %lf %lf", &n, &perimeter, &apothem) != 3) {
printf("Invalid Input\n");
return 1;
}
if (n < 3 || perimeter <= 0 || apothem <= 0) {
printf("Invalid Input: Sides must be >= 3 and dimensions positive.\n");
return 1;
}
double side_length = perimeter / n;
double area = 0.5 * perimeter * apothem;
double exterior_angle = 360.0 / n;
printf("Side Length: %.2f\n", side_length);
printf("Area: %.2f\n", area);
printf("Exterior Angle: %.2f degrees\n", exterior_angle);
return 0;
}
[Wrote 28 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-Q Previous ^B Back ^← Prev Word ^A Home
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^W Where Was M-W Next ^F Forward ^→ Next Word ^E End


kali@kali: ~/HDOC_Day6
Session Actions Edit View Help
(kali㉿kali)-[~]
$ mkdir HDOC_Day6
cd HDOC_Day6
(kali㉿kali)-[~/HDOC_Day6]
$ nano q1.c
(kali㉿kali)-[~/HDOC_Day6]
$ gcc q1.c -o q1 -lm
./q1
Enter base, height of triangular field, and radius of circular pond: 10 8 2
Triangle Area: 40.00
Pond Area: 12.57
Remaining Area: 27.43
(kali㉿kali)-[~/HDOC_Day6]
$ nano q2.c
(kali㉿kali)-[~/HDOC_Day6]
$ gcc q2.c -o q2
./q2
Enter number of sides (n), perimeter, and apothem: 6 30 4.33
Side Length: 5.00
Area: 64.95
Exterior Angle: 60.00 degrees
(kali㉿kali)-[~/HDOC_Day6]
$